DAgger and HG-DAgger · For a VR class

Teaching a robot by taking over in VR

How a robot learns from the moments a person grabs the controls: two papers, DAgger (2011) and HG-DAgger (2019), and what we did with them in the 2026 BEHAVIOR Challenge.

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Robots learn tasks by copying people

A person teleoperates the robot, in VR or with controllers, and every frame is recorded: what the cameras saw, what the hands did.

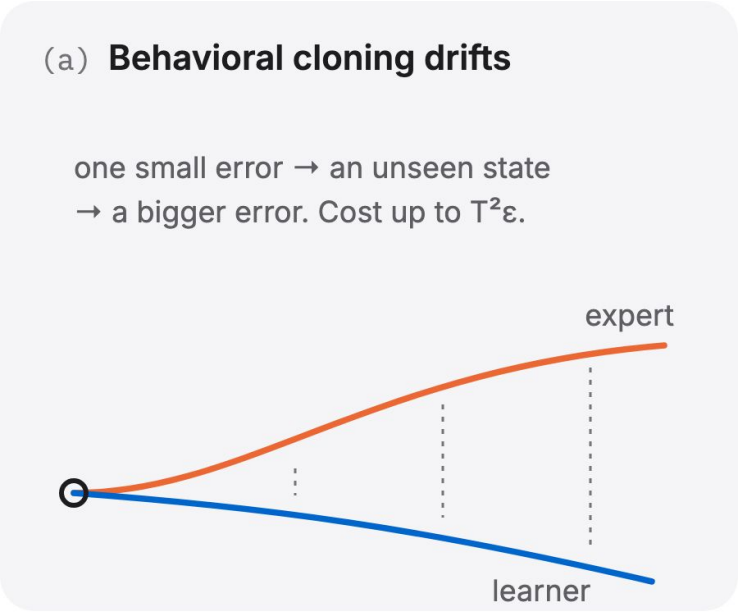
The BEHAVIOR Challenge provides 20,000 such demonstrations, 200 per task and 1,950 hours in all, for 100 household tasks in a simulated home.

A neural network is trained to copy them: images in, joint motions out. This is behavior cloning.



The robot's head and wrist cameras during one of our VR sessions. The red overlay means a person is driving.

A copy drifts: one small mistake leads to a place the demos never showed



The demonstrations show only what good operators do. They never show how to get back after drifting off course.

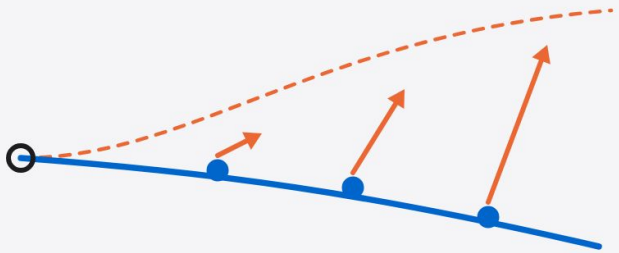
So the first small mistake puts the robot in a situation it has never seen. There it makes a bigger mistake, and so on.

Ross et al. (2011) proved it: over T steps, the damage can grow like T^2, not T.

DAgger: let the learner drive, and ask the expert what it would have done

(c) **DAgger labels the learner's states**

the learner drives; the expert says what it would do at each visited state

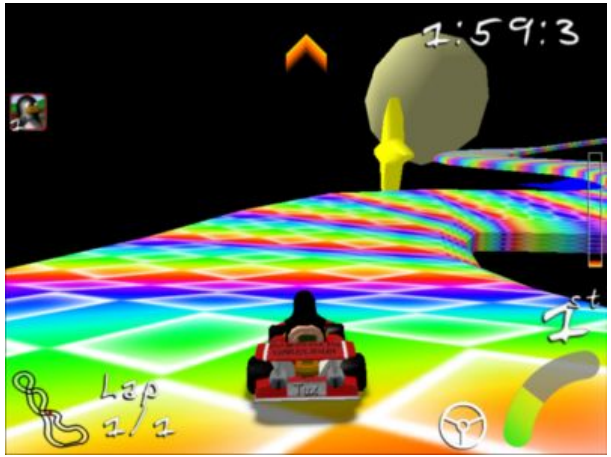


The learner drives. At every step, the expert says what it would do there. Those pairs are added to the data, the learner is retrained, and the loop repeats.

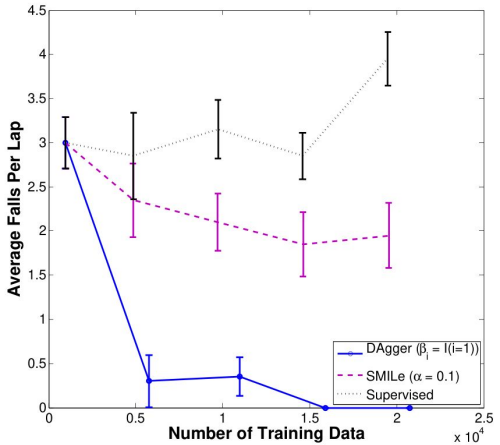
The data now covers the places the learner actually goes, and the damage grows like T again.

It works with any expert that can answer "what would you do here?": a person, a script, or a bigger model.

After 15 rounds of DAgger, the kart stopped falling off the track

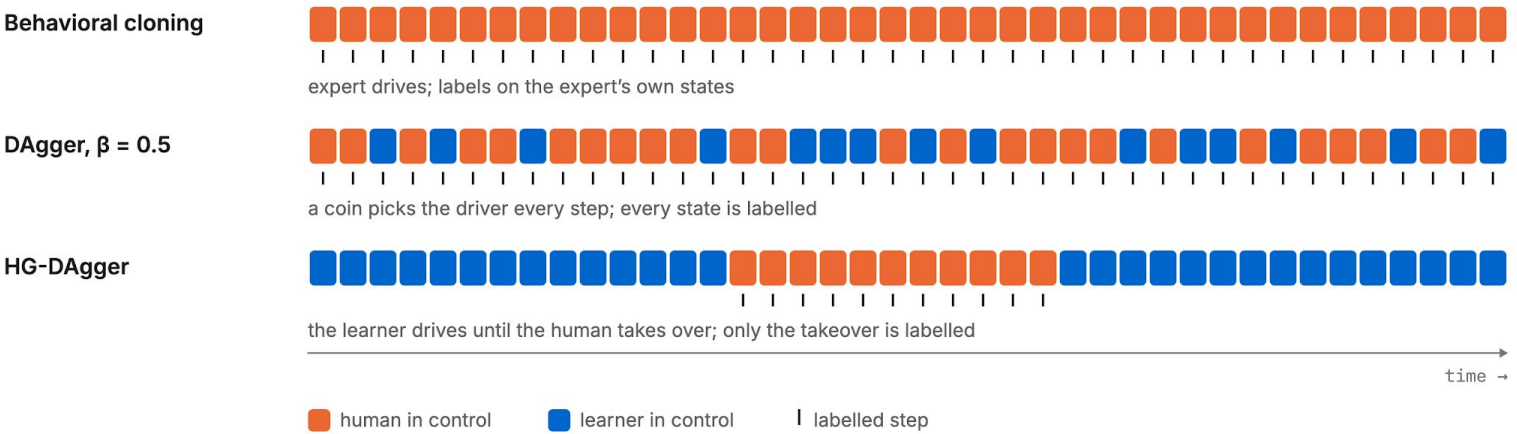


Super Tux Kart: steering from the screen image, a person as the expert (Ross et al., 2011).



Falls per lap against training data: DAgger reaches zero, plain copying never improves (Ross et al., 2011).

A person cannot label every step while the robot drives



Asking a person "what would you do here?" thirty times a second, without the controls, gives poor answers; mixing control step by step feels like lag. HG-DAgger changes who holds the controls, and when.

HG-DAgger: the human holds the gate

The robot drives. The person watches.

When it looks unsafe, the person takes over with full control, steers back to safety, and hands back.

Only the takeover is recorded as new training data.

Kelly et al. tested it on a real car with a second wheel in the passenger seat: no collisions in the first 5,000 samples, where plain copying and DAgger each had some.



The expert at an auxiliary wheel. Photo: Kelly et al., 2019 (HG-DAgger, Fig. 2).

In VR: our takeover loop

1. **Rewind**: the robot's own failed run is restored from a snapshot near the moment it went wrong.
2. **Watch**: you stand behind the robot in the headset while the policy drives.
3. **Take over**: press B. The arms follow your hands; the torso follows your height.
4. **Fix and hand back**: do the step as the demonstrations do, then press B again.



Streamed from the simulator into a Quest 3S or Pico 4 headset over NVIDIA CloudXR, control at 30 Hz. Each frame is tagged with who was driving.

The result: a training episode in the demonstrations' format, plus a per-frame "human" flag.

What we learned about people in the loop

- **People are slow**: 78 recoveries in four days, on one task, by three operators. The robot produced failures far faster than that.
- **Most failures are not fixable by a short takeover**: in 71% of them the robot never got going at all.
- **Prevention is not recovery**: resuming 5 s before the failure, people took over before it happened. 29 of 33 “place” recordings start with the can already in hand. We moved the resume point to 3 s after the mark.
- **The takeover is the data**: a clean per-frame “who is driving” flag and a clean hand-back matter as much as the steering itself.

When people ran out, scripts and a stronger policy took over

4,166

scripted recoveries on 48 tasks

1,028

clips watched by an AI reviewer

855 / 173

published / denied by a person

+7 points

pick-up success from a stronger policy's corrections

Per-skill scripts that read the simulator state recovered 3,436 of 8,303 failed phases. An AI agent watched each clip and wrote one line on it; a person made the final call.

The scripts' data hurt the policy (they moved a joint no person moves); a stronger policy's corrections helped.



A scripted recovery: chopping a sprout.

Where a VR designer makes the difference

- **The takeover control**: one button, full authority, instant hand-back. Shared control feels like lag.
- **What the operator sees**: the robot's cameras, the current step ("pick up the can"), and a clear who-is-driving cue (our red border).
- **Where the replay resumes**: too early, and people prevent instead of recover.
- **Latency**: control runs at 30 Hz over a streamed render, physics at 120 Hz; the hands must track without delay.
- **An open question**: HG-DAgger never measured how control authority and VR latency change the quality of a correction. A user study would.

Sources

1. Ross, Gordon and Bagnell. A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning. AISTATS 2011 (DAgger).
2. Kelly, Sidrane, Driggs-Campbell and Kochenderfer. HG-DAgger: Interactive Imitation Learning with Human Experts. ICRA 2019.
3. BEHAVIOR Challenge 2026, Stanford Vision and Learning Lab: 100 household tasks in OmniGibson.
4. Our team report, Harvesting Failures for Imitation (draft, November 2026): the VR loop, the 82 human recoveries, the scripted and policy experts, and the training results.
5. Images: DAgger paper (kart screenshot and chart), HG-DAgger paper (car photo), our own recordings (VR and chop stills). Drafted with AI tools; numbers checked against the papers and our records.